OSPF Finite State Machines

The OSPF Finite State Machine (FSM) describes the step-by-step process routers follow to form neighbor relationships and exchange routing information. OSPF doesn't just start sharing routes ~~asked~~ immediately, it goes through a series of well-defined states to ensure stability, accuracy, and synchronization between routers.

Think of it like a handshake process: before two routers trust each other and exchange full routing tables, they must pass through several states of verification and agreement.

The 7 OSPF Neighbor States

1. **Down State:**

This is the starting point. The router has no information about the neighbor yet. No OSPF Hello packets has been received.

This is the default state. Either the neighbor is not present or communication hasn't started

2. **Init State:**

The router has received a Hello packet from a neighbor, but:

- The ~~receiving~~ receiving router's own Router ID is not listed in that Hello Packet.
- This means communication is one-way at this stage

3. **2-Way State**

This is where things start getting important

- The router sees its own Router ID in the neighbor's Hello packet
- Communication is now bidirectional (two-way)
- This is the final state for some routers, especially on broadcast networks (like ethernet), where not all routers form full adjacencies.
- DR/BDR are elected at this state
- Only certain routers (DR/BDR) will move beyond this state on multi-access networks

4. **Ex Start State**

Now routers decide who will controll the exchange process. One router

becomes the Master, the other becomes the slave and the agree on initial sequence numbers for database exchange.

5. Exchange State:

At this state routers begin sharing summaries of their routing databases using

- Database Description (DBD) packet

Each router describes what it knows, but doesn't send full details yet.

6. Loading State

Router request missing information

- They send link-States Request (LSR) packets.
- Neighbors respond with Link-State Update (LSU) packets.
- Router acknowledge with LSAck Link-State Acknowledgment.

This continues until both routers have identical link-state databases

7. Full State.

This is the final and desired state.

- Routers are fully synchronized.
- They have identical Link-state Databases (LSDBs)
- Routing informations can now be reliably used to calculate shortest paths.

DR/BDR and FSM Behaviour

On multi-access networks (e.g., Ethernet):

- Not all routers form full adjacencies.
- Routers only fully synchronize with: Designated Router (DR)
  Backup Designated Router (BDR).

Other routers stay in the 2-way state with each other.

Understanding the OSPF FSM is crucial because:

- It helps in troubleshooting adjacency issues.
- You can identify where a problem occurs.
- It ensures reliable and loop-free routing